

Elements of Software Engineering

Nguyễn Thị Minh Tuyền · Faculty of Information Technology, HCMUS

Teaching Staff

- Instructor: Nguyễn Thị Minh Tuyền
 - FIT, HCMC University of Science
 - Email: ntmtuyen@fit.hcmus.edu.vn
- Teaching assistant:
 - Mai Anh Tuan (matuan@hcmus.edu.vn)
 - Ho Tuan Thanh (htthanh@hcmus.edu.vn)

Course Description

- This course provides a comprehensive introduction to the principles and practices of software engineering.
- Students will explore key concepts such as software development processes, agile methodologies, and the fundamental activities involved in building software – from initial requirements and design to implementation, testing, and maintenance.
- Through team-based projects, students will apply software engineering techniques to develop real-world applications collaboratively.

Topics covered

1. Overview + Introduction to LLMs, Prompting & Vibe Coding
2. Software processes
3. Agile software development + SDD, GitHub Spec Kit
4. Software requirements engineering
5. System modeling
6. Architectural design
7. Design and implementation
8. User interface design
9. Software testing
10. Project management + Review

Learning Outcomes

- **After completing this course, students will be able to:**
 - Understand fundamental concepts, principles, methods, and techniques in software engineering.
 - Apply requirements engineering concepts to define and document system requirements.
 - Analyze and design software systems using appropriate modeling techniques.
 - Design simple and effective user interfaces.
 - Develop test plans and test cases for software systems.
 - Apply software testing techniques to ensure software quality.
 - Select and justify a suitable software development process based on project characteristics.
 - Collaborate effectively and demonstrate teamwork skills in software development projects.

References

- **Software Engineering**, 10th edition, Ian Sommerville, Addison-Wesley, 2015 (Textbook)
- **Software Engineering: A Practitioner's Approach**, 9th edition, Roger S. Pressman, McGraw-Hill Higher Education, 2019
- **Software Engineering with UML**, Bhuvan Unhelkar, CRC Press, 2018

Grading

- Project 40%
- Classwork 10%
- 2 Tests (W6, W10) 15%
- Final exam 35%

Please note that delivering the final oral project presentation is a mandatory requirement to pass the course.

Project

- **Project assignments**

- Students will work in teams of five to complete a comprehensive software development project.
- Each team will perform all major activities of the software development lifecycle, from requirements analysis and system design to implementation, testing, and deployment.
- Teams are required to deliver both written documentation and oral presentations to demonstrate their project progress and outcomes.

Course Requirements

- **Class Participation and Communication**

- Students are encouraged to actively engage by asking questions during class sessions, through the online discussion forum, via email, or in person.

- **Late Submission Policy**

- A 15% grade deduction will be applied for each day an assignment is submitted late.
- Submissions that are four or more days late will receive a grade of zero.
- Exceptions may be granted in special circumstances (e.g., illness or emergencies) with appropriate documentation and prior communication with the instructor.

Academic Integrity and Attendance Policies

- **Attendance**
 - Students may not be absent for more than 30% of total class sessions. Exceeding this limit will result in being ineligible to take the test or final examination.
 - Students are expected to arrive on time and remain until the end of each class session.
- **Academic Integrity:** Academic honesty is essential. Cheating or plagiarism in any form is strictly prohibited.
 - Students may not copy from classmates, friends, or online sources without proper citation.
 - Students may not allow others to copy their work.
 - Any violation of these rules will result in a grade of zero (0) for the entire course.

Questions?

Nguyễn Thị Minh Tuyền · Faculty of IT, HCMUS

CS300 · Elements of Software Engineering · Week 1